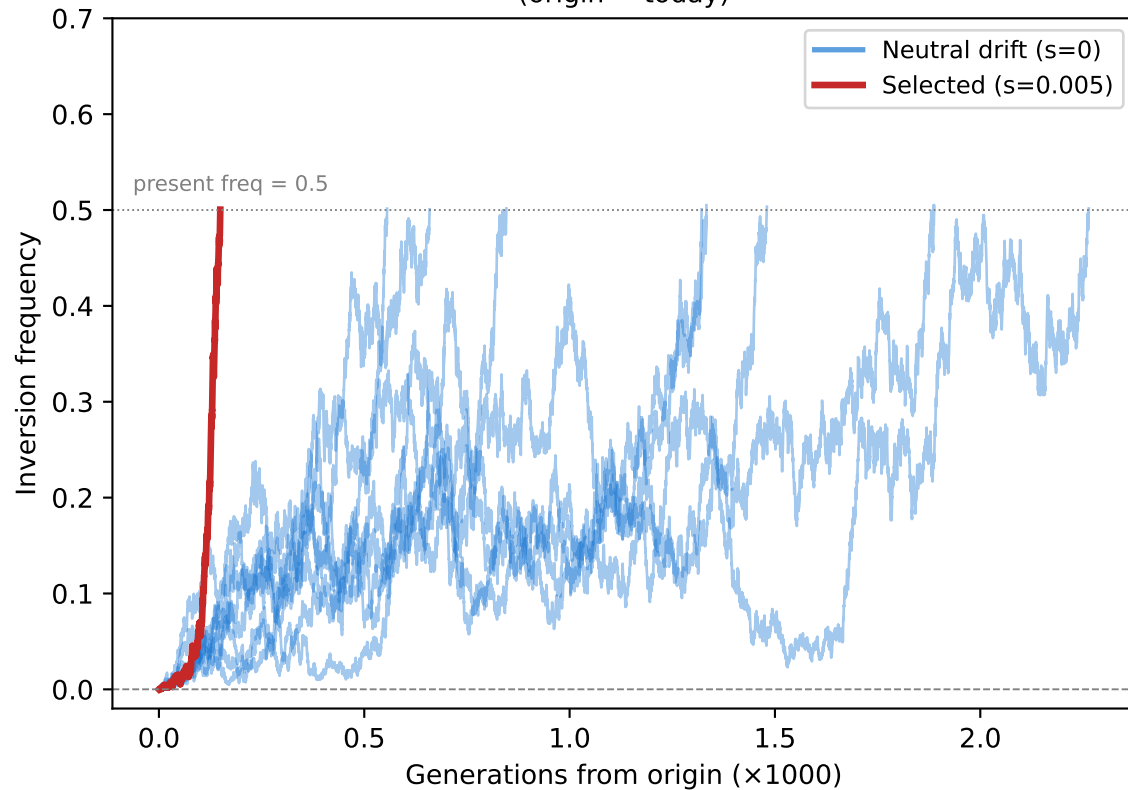


A. Forward-in-time WF trajectories
(origin $\rightarrow$ today)



B. Inversion age varies widely
(same present freq, different histories)

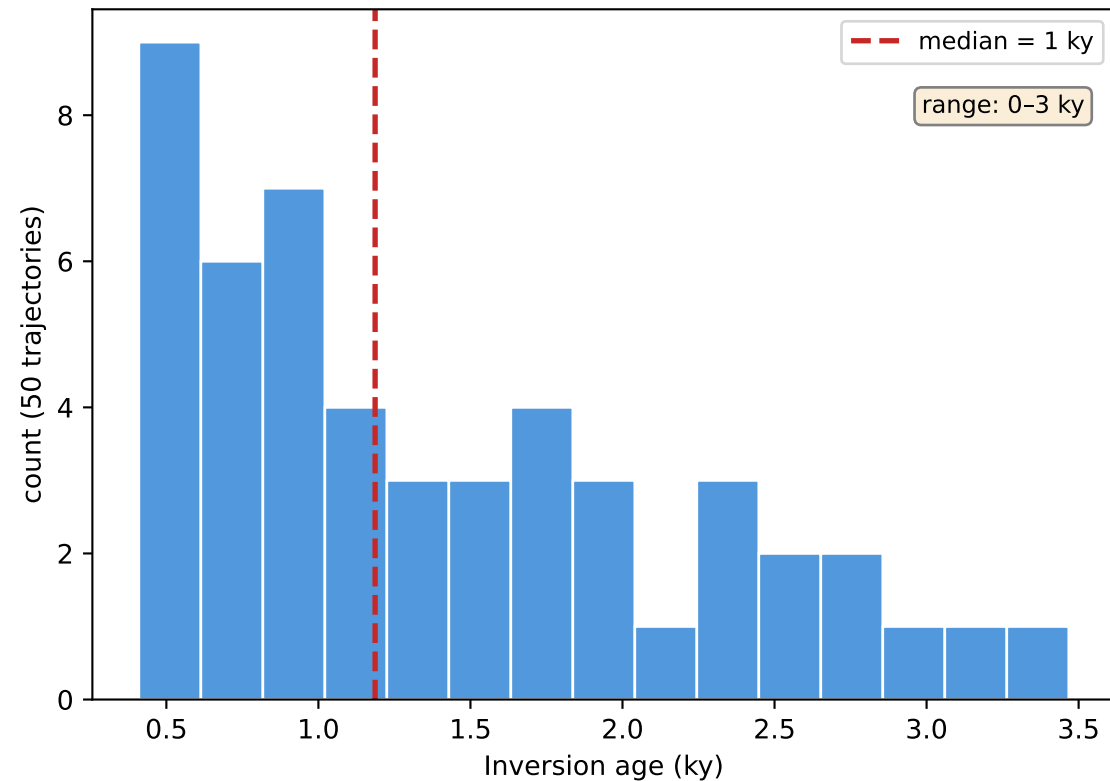


Figure 5. Forward-in-time Wright-Fisher trajectories for inversion frequency. (A) Example trajectories from $1/(2N)$ to a present frequency of 0.5. Neutral drift (blue, 8 runs) takes highly variable paths; positive selection ($s=0.005$, red) reaches the same frequency much faster. (B) Distribution of inversion ages (time from origin to reaching $p=0.5$) across 50 neutral trajectories. Even for the same present frequency, inversion ages vary by orders of magnitude — this motivates ABC inference rather than point estimates of t_{inv} . Parameters: $N_e=10,000$, 10 gen/yr, $p_{\text{target}}=0.5$.
Command: `pixi run -e all python examples/make_figures.py`